

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4_AT2 — VISUALIZATION EVALUATION EXERCISE

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO4 – Evaluation (BL5)	Assessment:	Assessment Tool 2 – Visualization Evaluation Exercise
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO4 Statement:	<i>Evaluate the appropriateness and design choices of relationship-visualization techniques (scatterplots, correlograms, PCA, bubble/hexbin plots, and network graphs) and justify improvements.(BL5)</i>		
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Section / Batch:	SLOT C	Date:	22/08/2026

Code of Conduct

I JOTHIKA M (192324235) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *Jothika M*

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must go beyond describing the scenario: evaluate the specific strength or weakness, and justify your judgment.
- Where a question asks for a recommendation, propose a specific alternative — not just “use a different chart.”
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – VISUALIZATION EVALUATION EXERCISE

Problem 2 (5 Marks)

Data: An analyst plots 50,000 overlapping points on a scatterplot to show a relationship, resulting in a single solid blob.

Task: Evaluate this choice and recommend a more appropriate alternative.

Problem 9 (5 Marks)

Data: A correlogram omits the diagonal cells (each variable correlated with itself).

Task: Evaluate whether this omission represents a meaningful loss of information.

Problem 11 (5 Marks)

Data: A team applies PCA to a dataset where original features are on very different scales (e.g., income in dollars, age in years) without standardizing them first.

Task: Evaluate the risk this creates for the resulting principal components.

Problem 18 (5 Marks)

Data: A hexbin plot is used to visualize a dataset containing only 50 data points.

Task: Evaluate whether hexbin binning was necessary or appropriate for this dataset size.

Problem 25 (5 Marks)

Data: A network graph's layout algorithm places two clearly separate, unconnected clusters directly on top of each other visually.

Task: Evaluate this rendering issue and its impact on how the graph would be interpreted.

Rubrics for Calculation **ASSESSMENT RUBRIC**
AT2 — Visualization Evaluation Exercise

CO4 · BT5: Evaluate ·

This rubric is used to assess student responses to the CO4-AT2 Visualization Evaluation Exercise problems, in which students are given a visualization design choice and must evaluate its appropriateness, identify limitations, and justify improvements.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of the relevant visualization principle being evaluated (e.g., overplotting, PCA variance, encoding choice, network readability).	Good understanding of the relevant principle, with only minor gaps.	Basic understanding, but with some misconceptions about which principle applies.	Poor understanding of core principles; misidentifies the nature of the issue entirely.
Explanation	25%	Clearly explains why the design choice is effective or ineffective, including its practical consequence for a reader — not just that it is ‘good’ or ‘bad.’	Explains the evaluation with adequate reasoning, covering most of the practical consequence.	Identifies the issue but explanation is generic or only loosely tied to its practical effect.	Little or no explanation provided; the evaluation is a bare assertion.
Application	20%	Proposes a specific, well-justified recommendation or alternative that directly addresses the evaluated issue.	Proposes a workable recommendation, with only minor gaps in justification.	Proposes a recommendation, but it only partially addresses the issue or lacks justification.	Fails to propose a workable recommendation, or it does not address the evaluated issue.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Organization	15%	The response is clearly structured, addressing evaluation and recommendation in a logical sequence with coherent overall flow.	The response is organized, with only minor issues in flow or clarity.	Basic structure; both parts are addressed but the response lacks clarity or sequence.	Poorly organized; the response is incoherent, and one or both parts are left unaddressed.
Accuracy	10%	Both the evaluation and the recommendation are completely accurate and well-suited to the specific scenario described.	Mostly accurate, with only minor omissions or a small error.	Partially correct; the evaluation or recommendation contains notable inaccuracies.	Largely incorrect or inappropriate, with major gaps in both the evaluation and recommendation.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 2 – Scatterplot with 50,000 Overlapping Points

Introduction

A scatterplot is commonly used to visualize the relationship between two numerical variables. Each observation is represented by an individual point. However, when a very large number of observations are plotted in the same space, the points may overlap heavily. This is known as **overplotting**.

In this case, **50,000 overlapping points form a single solid blob**, so the original scatterplot is not an effective way to communicate the relationship.

Evaluation

The choice of using a normal scatterplot for 50,000 observations is **not appropriate when severe overplotting occurs**.

The major problem is that thousands of points are drawn on top of one another. As a result:

- Individual observations cannot be distinguished.
- Dense and sparse regions cannot be easily identified.
- The underlying relationship between the variables becomes difficult to see.
- Clusters may be hidden.
- Outliers can also become difficult to identify.
- The chart contains a large amount of data but communicates very little useful information.

Therefore, the problem is not simply that there are 50,000 observations; the problem is that **all observations are being represented as individual overlapping marks**.

Recommended Alternative

A **hexbin plot** is a more appropriate alternative.

A hexbin plot divides the plotting area into hexagonal regions and counts the number of observations falling inside each region. The density of observations can then be represented using shading.

Another suitable option is a **2D density plot**.

Why This Is Better

The alternative summarizes the data while preserving the important distributional information.

It allows the reader to identify:

- High-density areas
- Low-density areas
- Clusters
- General relationships
- Possible unusual regions

Thus, instead of seeing one solid blob, the reader can see the **structure of the relationship**.

R Code – Hexbin Plot

```
R
# Install once if required
# install.packages("ggplot2")

library(ggplot2)

# Generate example data
set.seed(123)

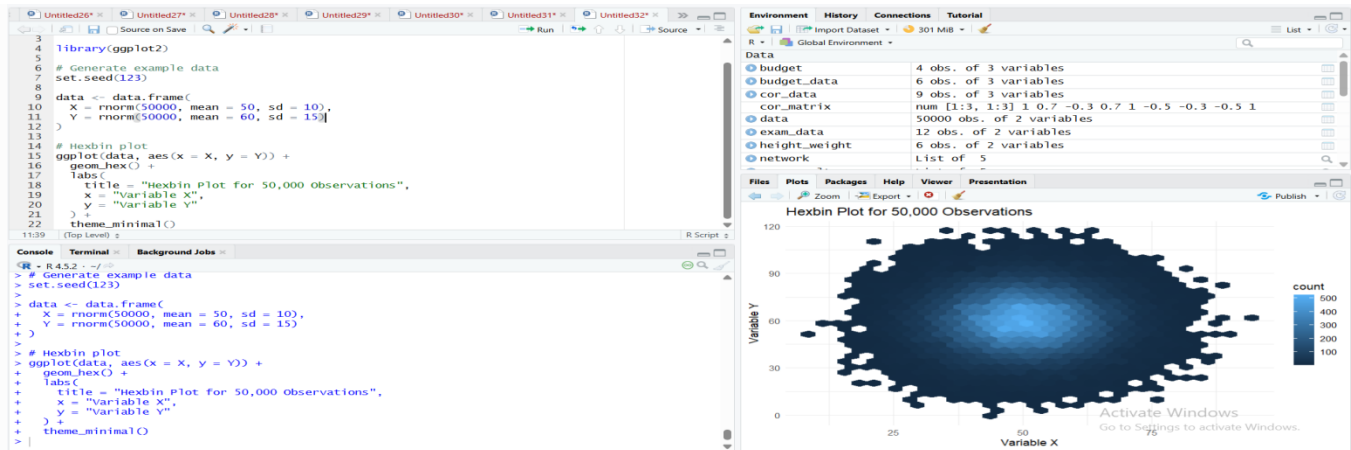
data <- data.frame(
  X = rnorm(50000, mean = 50, sd = 10),
  Y = rnorm(50000, mean = 60, sd = 15)
)

# Hexbin plot
ggplot(data, aes(x = X, y = Y)) +
  geom_hex() +
  labs(
    title = "Hexbin Plot for 50,000 Observations",
```

```

x = "Variable X",
y = "Variable Y"
) +
theme_minimal()

```

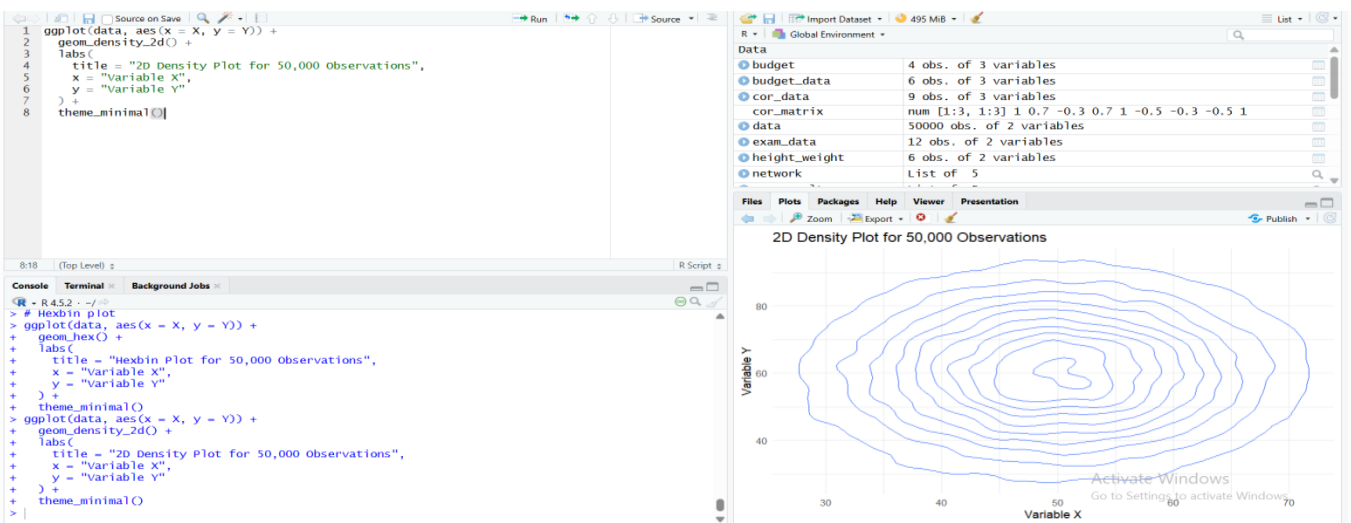


Alternative – 2D Density Plot

```

R
ggplot(data, aes(x = X, y = Y)) +
  geom_density_2d() +
  labs(
    title = "2D Density Plot for 50,000 Observations",
    x = "Variable X",
    y = "Variable Y"
  ) +
  theme_minimal()

```



Conclusion

A standard scatterplot is inappropriate when 50,000 points overlap into a solid blob because the underlying pattern becomes invisible. A **hexbin or 2D density plot** is more appropriate because it summarizes point density and makes the relationship easier to interpret.

Problem 9 – Correlogram Without Diagonal Cells

Introduction

A **correlogram** represents the correlation between pairs of variables. The diagonal cells of a correlation matrix represent the correlation of every variable with itself.

The correlation of a variable with itself is always:

$$r=1$$

Evaluation

Omitting the diagonal cells does **not represent a meaningful loss of information**.

For example:

	Variable A	Variable B	Variable C
Variable A	1.00	0.70	-0.30
Variable B	0.70	1.00	-0.50
Variable C	-0.30	-0.50	1.00

The diagonal contains:

- A with A = 1
- B with B = 1
- C with C = 1

These values are automatic and provide no new information about relationships **between different variables**.

Why the Omission Is Acceptable

Removing the diagonal can actually make the correlogram cleaner because the reader can focus on the meaningful pairwise correlations.

Also, a correlation matrix is symmetrical:

$$r_{AB}=r_{BA}$$

Therefore, even one half of the matrix can sometimes be displayed.

Important Qualification

The omission is acceptable **provided that the chart is clearly labeled or explained.**

If the diagonal is removed without any indication, an inexperienced reader might think data are missing. A short note such as:

“Diagonal correlations are omitted because each variable correlates perfectly with itself ($r = 1$).”

would eliminate this ambiguity.

R Code

```
R
library(ggplot2)

# Correlation matrix
cor_matrix <- matrix(
  c(
    1.0, 0.7, -0.3,
    0.7, 1.0, -0.5,
    -0.3, -0.5, 1.0
  ),
  nrow = 3,
  byrow = TRUE
)

colnames(cor_matrix) <- c(
  "Rainfall",
```



```

"Crop Yield",
"Temperature"
)

rownames(cor_matrix) <- colnames(cor_matrix)

# Convert matrix to data frame
cor_data <- as.data.frame(as.table(cor_matrix))

colnames(cor_data) <- c(
  "Variable1",
  "Variable2",
  "Correlation"
)

# Remove diagonal
cor_data <- cor_data[
  cor_data$Variable1 != cor_data$Variable2,
]

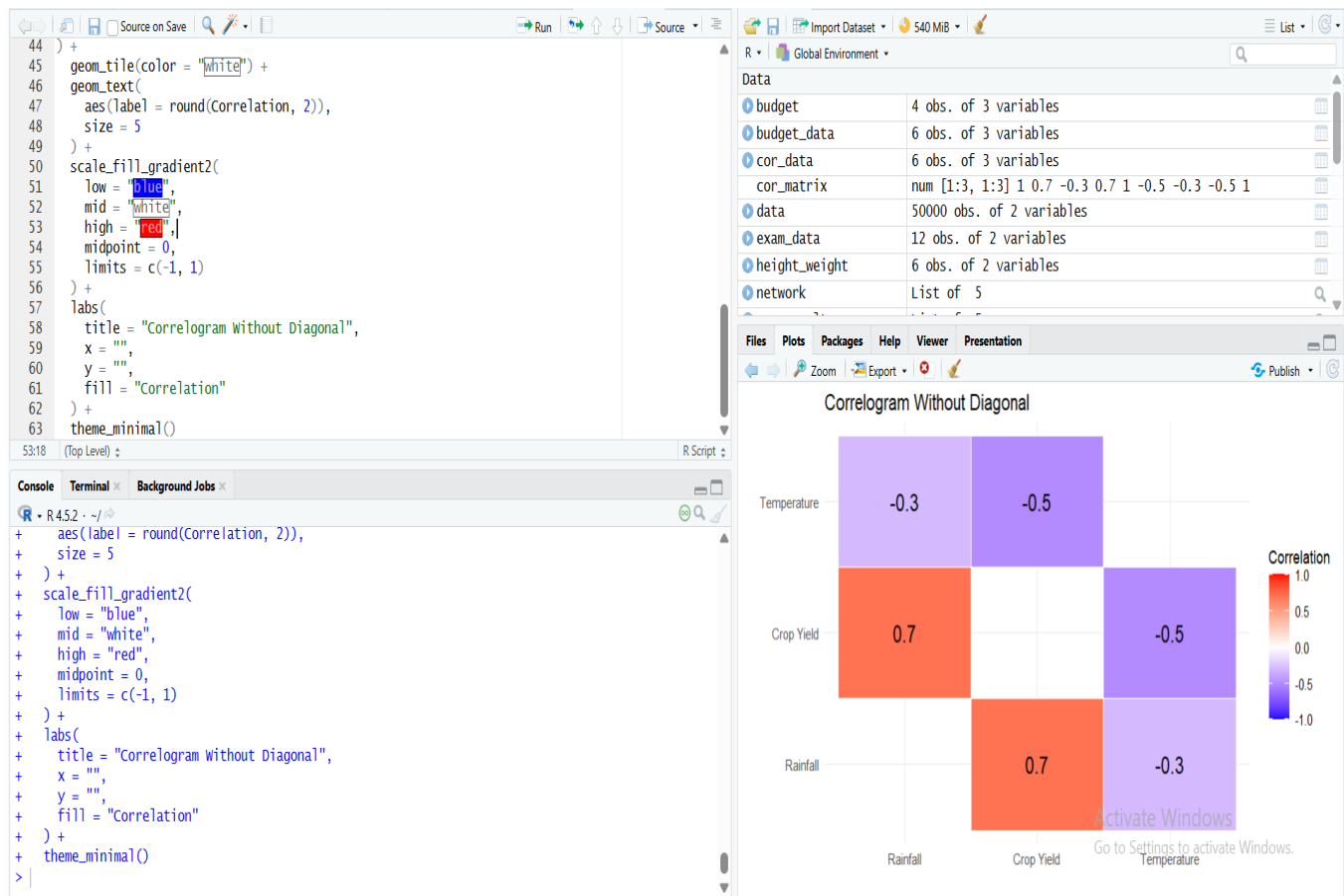
# Draw correlogram
ggplot(
  cor_data,
  aes(
    x = Variable1,
    y = Variable2,
    fill = Correlation
  )
) +
  geom_tile(color = "white") +
  geom_text(
    aes(label = round(Correlation, 2)),
    size = 5
  ) +
  scale_fill_gradient2(
    low = "blue",
    mid = "white",
    high = "red",
    midpoint = 0,
    limits = c(-1, 1)
  ) +
  labs(

```

```

title = "Correlogram Without Diagonal",
x = "",
y = "",
fill = "Correlation"
) +
theme_minimal()

```



Conclusion

Omitting the diagonal cells is **appropriate** and **does not cause meaningful information loss**, because each diagonal value is always +1. The omission can reduce unnecessary visual clutter, provided that the reason for the omission is clearly communicated.

Problem 11 – PCA Without Standardization

Introduction

Principal Component Analysis (PCA) is a dimensionality-reduction technique that creates new variables called principal components. These components are formed to capture the maximum variation in the original variables.

However, PCA is sensitive to the **scale of the input variables**.

Scenario

Suppose a dataset contains:

- Income → measured in dollars
- Age → measured in years

These variables are measured on very different numerical scales.

Evaluation

Applying PCA without standardizing the variables creates a **significant risk that variables with larger numerical variance will dominate the principal components**.

For example:

Income → 20,000 – 100,000
Age → 18 – 60

The numerical variation in income is much larger than the numerical variation in age.

Consequently, PCA may give income much greater influence simply because of its measurement scale.

Problems Created

Without standardization:

1. PC1 may be dominated by income.
2. Age may contribute relatively little.
3. The components may reflect measurement units rather than meaningful relationships.
4. Important information from smaller-scale variables may be underrepresented.

5. Interpretation of the principal components may become misleading.

Therefore, the resulting PCA may not fairly represent the contribution of the original features.

Recommended Solution

Standardize the variables before performing PCA.

The standardization formula is:

$$Z = \frac{X - \bar{X}}{SD}$$

After standardization, each variable has approximately:

- Mean = 0
- Standard deviation = 1

This puts the variables on a comparable scale.

R Code – Without Standardization

```
R
data <- data.frame(
  Income = c(30000, 45000, 60000, 75000, 90000),
  Age = c(22, 30, 35, 42, 50)
)

# PCA without standardization
pca_wrong <- prcomp(
  data,
  scale. = FALSE
)

print(pca_wrong)

summary(pca_wrong)
```

```
Standard deviations (1, ..., p=2):  
[1] 2.371708e+04 7.745966e-01
```

```
Rotation (n x k) = (2 x 2):
```

	PC1	PC2
Income	0.9999998972	0.0004533333
Age	0.0004533333	-0.9999998972

```
>
```

```
> summary(pca_wrong)
```

```
Importance of components:
```

	PC1	PC2
Standard deviation	23717	0.7746
Proportion of Variance	1	0.0000
Cumulative Proportion	1	1.0000

R Code – Recommended PCA

```
R
```

```
# PCA with standardization
```

```
pca_correct <- prcomp(  
  data,  
  scale. = TRUE  
)
```

```
print(pca_correct)
```

```
summary(pca_correct)
```

```
Standard deviations (1, ..., p=2):  
[1] 1.41329930 0.05084385
```

```
Rotation (n x k) = (2 x 2):
```

	PC1	PC2
Income	0.7071068	0.7071068
Age	0.7071068	-0.7071068

```
>
```

```
> summary(pca_correct)
```

```
Importance of components:
```

	PC1	PC2
Standard deviation	1.4133	0.05084
Proportion of Variance	0.9987	0.00129
Cumulative Proportion	0.9987	1.00000

The important change is:

```
R  
scale. = TRUE
```

Interpretation

Standardization prevents the variable with the largest numerical scale from automatically dominating the PCA.

After standardization, PCA focuses more appropriately on the **patterns of variation and relationships among the variables**, rather than their measurement units.

Conclusion

Applying PCA without standardizing features measured on very different scales can produce **scale-dominated and misleading principal components**. Standardization should generally be performed before PCA when the variables are measured in substantially different units.

Problem 18 – Hexbin Plot with Only 50 Data Points

Introduction

A **hexbin plot** groups observations into hexagonal bins and represents the number of observations within each bin. It is particularly useful when a scatterplot contains many points and suffers from overplotting.

Here, however, the dataset contains only **50 observations**.

Evaluation

Using a hexbin plot for only 50 data points is generally **not necessary**.

With 50 observations, a normal scatterplot can usually display the individual points clearly.

Using hexbinning in this situation can introduce unnecessary aggregation:

- Individual observations are hidden inside bins.
- Exact point locations are less visible.
- The visualization becomes more complicated than necessary.
- Binning may obscure small patterns or individual unusual observations.

Therefore, the hexbin technique is excessive for this dataset size unless there is a specific reason that the 50 points overlap heavily.

Recommended Alternative

Use a **standard scatterplot**.

If a few observations overlap, use:

- Transparency
- Jitter
- Point labels where necessary

This preserves the individual observations while remaining easy to read.

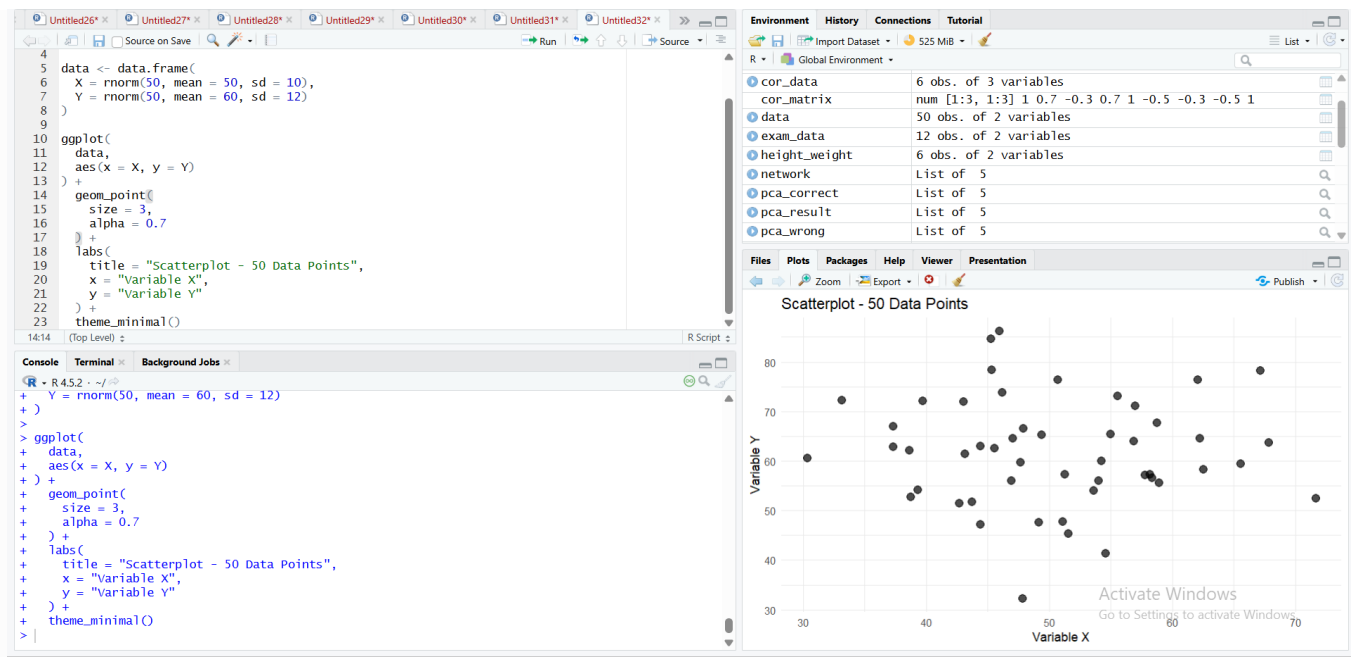
R Code – Recommended Scatterplot

```
R
library(ggplot2)

set.seed(123)

data <- data.frame(
  X = rnorm(50, mean = 50, sd = 10),
  Y = rnorm(50, mean = 60, sd = 12)
)

ggplot(
  data,
  aes(x = X, y = Y)
) +
  geom_point(
    size = 3,
    alpha = 0.7
  ) +
  labs(
    title = "Scatterplot - 50 Data Points",
    x = "Variable X",
    y = "Variable Y"
  ) +
  theme_minimal()
```



For this scenario, the **scatterplot** is the more appropriate choice.

Interpretation

The purpose of visualization should determine the level of aggregation. Since only 50 observations are available, displaying them individually gives the reader more information than grouping them into bins.

Hexbinning becomes more useful when a much larger number of observations creates substantial overplotting.

Conclusion

Hexbinning is **generally unnecessary for a dataset of only 50 observations**. A standard scatterplot is simpler and preserves individual observations more effectively.

Problem 25 – Overlapping Network Clusters

Introduction

A network graph represents entities as nodes and their relationships as edges. A layout algorithm determines where these nodes are placed visually.

In this scenario, there are **two separate, unconnected clusters**, but the layout places them directly on top of each other.

Evaluation

This is a **serious rendering/layout problem** because the visual position of the nodes can create a false impression of connectivity.

The two clusters are actually unconnected, but their overlap can make the reader believe that:

- There are edges between the clusters.
- The clusters form one connected component.
- Certain nodes act as bridges.
- Relationships exist where none actually exist.

This means the visualization can communicate a structure that is different from the actual network data.

Why It Is Misleading

Consider the actual network:

Cluster 1	Cluster 2
A — B	X — Y
C — D	Z — W

These are two independent components.

If the layout overlaps them:

A/X	
/	\
B/Y	C/Z

\ /
D/W

The viewer may incorrectly interpret the proximity as a relationship.

Recommended Redesign

The two connected components should be **spatially separated**.

A suitable network layout should:

- Place separate clusters in different areas.
- Avoid node overlap.
- Reduce edge crossings.
- Provide sufficient spacing.
- Preserve the actual connectivity structure.

For example:



Now the separation makes the lack of connections immediately visible.

R Code

If igraph is not installed:

```
R  
install.packages("igraph")
```

Then:

```
R  
library(igraph)
```

```
# Two separate clusters  
edges <- c(  
  "A", "B",  
  "B", "C",  
  "C", "A",
```

```

    "X", "Y",
    "Y", "Z",
    "Z", "X"
  )

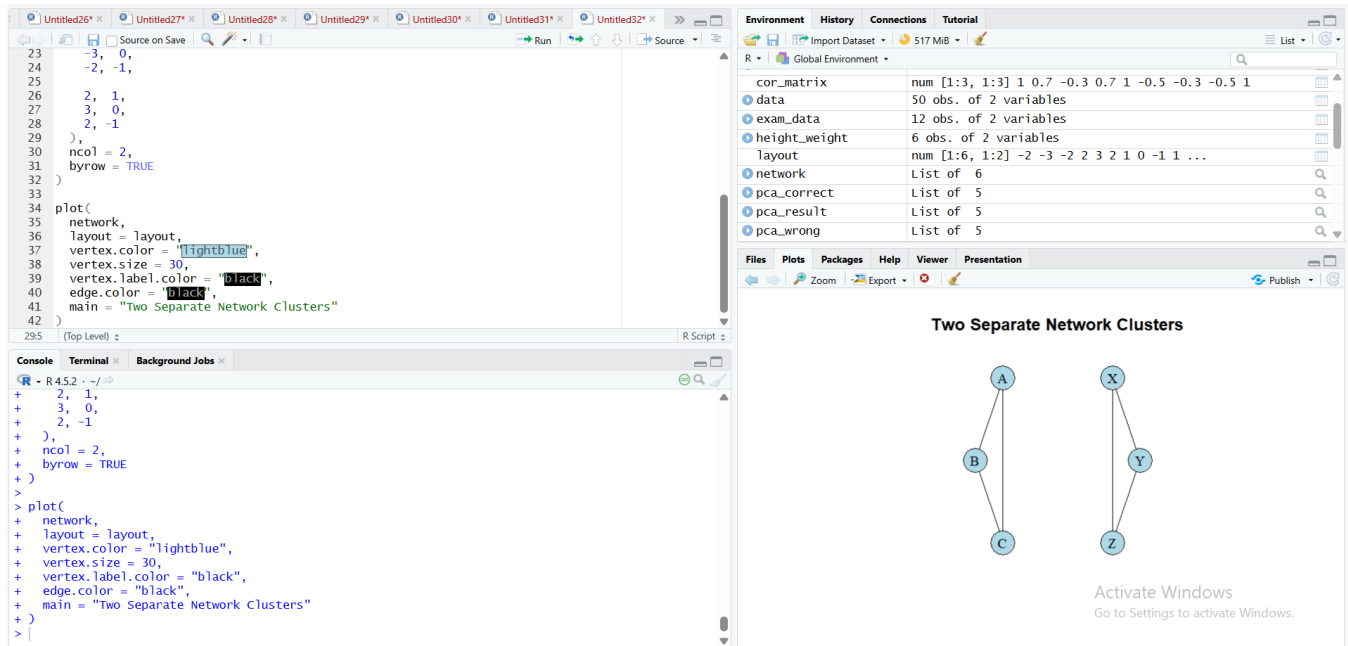
network <- graph(
  edges,
  directed = FALSE
)

# Manual layout to keep clusters separate
layout <- matrix(
  c(
    -2, 1,
    -3, 0,
    -2, -1,

    2, 1,
    3, 0,
    2, -1
  ),
  ncol = 2,
  byrow = TRUE
)

plot(
  network,
  layout = layout,
  vertex.color = "lightblue",
  vertex.size = 30,
  vertex.label.color = "black",
  edge.color = "black",
  main = "Two Separate Network Clusters"
)

```



Interpretation

The graph contains two independent connected components. The layout should make this structural fact visually obvious.

The problem is not with the network data itself; it is with the **rendering/layout**. A good network visualization should avoid making spatial overlap look like a connection.

Conclusion

Placing two unconnected clusters directly on top of each other is misleading because it can create **false visual connections**. The layout should separate the components and minimize node and edge overlap so that the true network structure is immediately understandable.